Unit IV: Electromagnetic Induction and Alternating Currents

CHAPTER–7: ALTERNATING CURRENT

GIST OF CHAPTER:

Alternating currents, peak and RMS value of alternating current/voltage; reactance and impedance; LCR series circuit (phasors only), resonance, power in AC circuits, power factor, wattles current. AC generator, Transformer.

<u>CONTENT/ CONCEPTS-</u>

<u>AC VOLTAGE AND AC CURRENT</u>: -The voltage and current whose magnitude changes continuously and polarity / direction reverses periodically is called alternating voltage and alternating current.

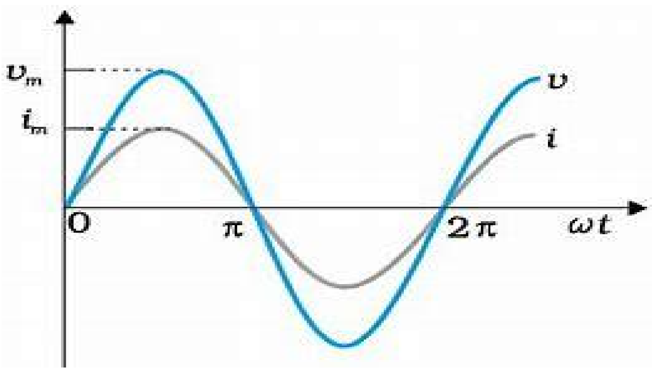
DC source supplies steady unidirectional current and controlled by ohmic resistances. Apart from ohmic resistance AC can also be controlled by inductors and capacitors.

AC is preferred over DC mainly because of the following reasons-

(a) Long distance transmission of electrical energy in the form of AC is economical.
(b) AC voltage can step or step down easily with the help of transformers.

ALTERNATING CURRENT	ALTERNATING VOLTAGE
$I = I_m \sin \omega t$	$V = V_m \sin \omega t$
I→instantanious value of current I_m →Peak value of current ω→ angular freq. rad/s ϕ→ phase angle, it gives information about the variation of alternating current with respect to the alternating voltage.	V →instantanious value of voltage V_m →Peak value of voltage ω → angular freq. rad/s

<u>VARIATION OF AC VOLTAGE AND AC CURRENT WITH TIME</u>: -



Average or mean value of AC over a cycle: - In 1 cycle of an AC it has positive values in the first half of a cycle and equal negative values in the second half of a cycle. Average becomes zero over a cycle. Hence, Average value is measured over half of a cycle and its value is $\boldsymbol{I_{av}}=\frac{\boldsymbol{2I_0}}{\boldsymbol{\pi}}$.

To measure the alternating voltage or current heating effect of current is used which is independent of the direction, that value is called as R.M.S or Effective or Virtual value.

R.M.S or Effective or Virtual value: - R.M.S value of A.C is defined as the D.C equivalent which produces the same amount of heat energy in same time as that of an A.C when passed through the same resistor.

Relation between R.M.S. value and peak value is

$$I_{rms} = \frac{I_m}{\sqrt{2}} \ \ and\ V_{rms} = \frac{V_m}{\sqrt{2}}$$

AC VOLTAGE APPLIED TO A RESISTOR

Consider an A.C voltage,

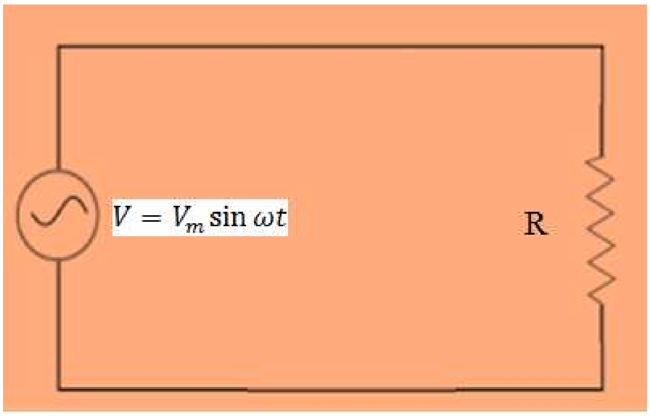
$$V = V_m \sin \omega t \text{..................} (1)$$

Divide both the side of the equation by R ,

$$\Rightarrow \frac{V}{R} = \frac{V_m}{R}\sin \omega t$$

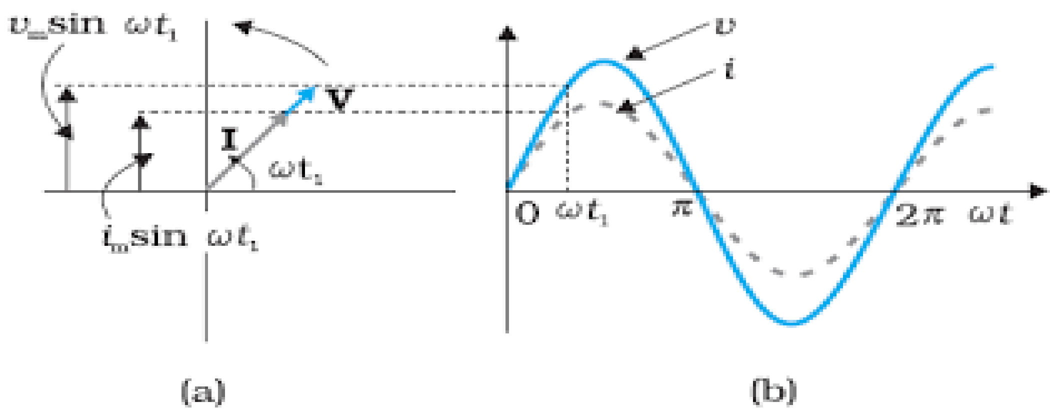
$$\Rightarrow I = I_m \sin \omega t \text{ ....} (2)$$

$$where\ I_m = \frac{V_m}{R}$$

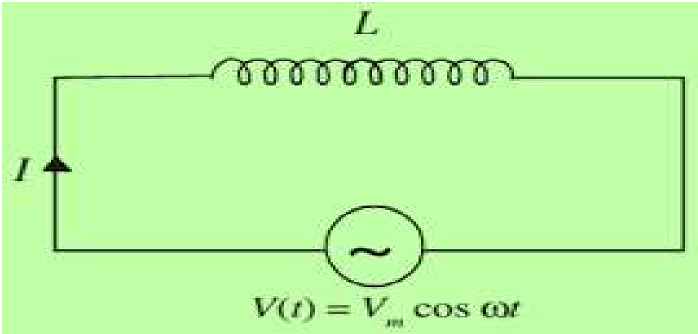


From equation (1) and (2) we can see that both the **voltage and current are in phase** with each other.

Phasor Diagram for pure resistive circuit: -



AC VOLTAGE APPLIED TO AN INDUCTOR: -



Consider an A.C voltage,

$V = V_m \sin \omega t$………………. (1)

Using the Kirchhoff's loop rule,

$\Rightarrow V - L\dfrac{dI}{dt} = 0$

$\Rightarrow V_m \sin \omega t - L\dfrac{dI}{dt} = 0$

$\Rightarrow \int dI = \dfrac{V_m}{L}\int \sin \omega t \, dt \Rightarrow I = \dfrac{V_m}{\omega L}\,(-\cos \omega t)$

$\Rightarrow I = \dfrac{V_m}{\omega L}\sin(\omega t - \dfrac{\pi}{2}) = I_m \sin(\omega t - \dfrac{\pi}{2}) \qquad where\ I_m = \dfrac{V_m}{\omega L}$

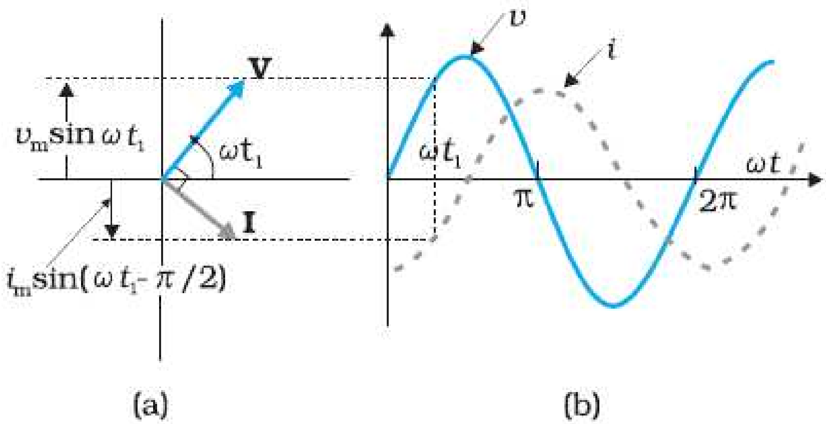
Thus, a comparison of equations for the source voltage and the current in an inductor shows that the current lags the voltage by π/2 or one-quarter (1/4) cycle. Now,

$I = I_m \qquad when\ \sin(\omega t - \dfrac{\pi}{2}) = 1$

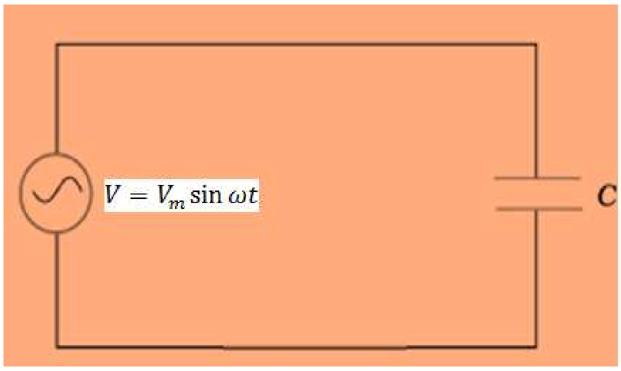
$\Rightarrow \quad \boxed{i_m = \dfrac{v_m}{X_L}}$

- **X_L= ωL** it is called as inductive reactance. This is the resistance offered by the inductor toan ac through it. It offers no resistance to a dc current.

Phasor Diagram for pure inductive circuit:



AC VOLTAGE APPLIED TO A CAPACITOR



Consider an A.C voltage,

$V = V_m \sin \omega t$.................. (1)

The instantaneous voltage V across the capacitor is

$$V = \frac{q}{C} - - - - - - - - - - - (2)$$

From (1) and (2) we get

$$V_m \sin \omega t = \frac{q}{C}$$

If I is the instantaneous current flowing through the capacitor,

$$I = \frac{dq}{dt} = \frac{d(CV_m sin\omega t)}{dt} = CV_m\omega \ \cos \omega t = \frac{V_m}{\frac{1}{\omega C}} \sin\left(\omega t + \frac{\pi}{2}\right)$$

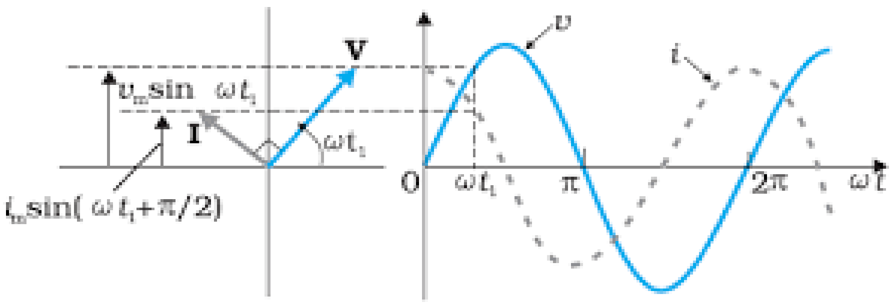
$$I = I_m \ \sin(\omega t + \frac{\pi}{2}) \qquad where \ I_m = \frac{V_m}{1/\omega C}$$

$$I = I_m \qquad when \ \sin(\omega t + \frac{\pi}{2}) = 1$$

In an ac circuit containing capacitor current leads the voltage by π/2 or voltage lags the current by π/2.

$X_C = \frac{\mathbf{1}}{\boldsymbol{\omega C}}$ = Capacitive reactance, This is the resistance offered by the capacitor to an AC through it it offers infinite resistance to dc current.

Phasor Diagram for pure capacitive circuit:-



Phasor diagram Graph of v vs ωt

For a purely capacitive circuit

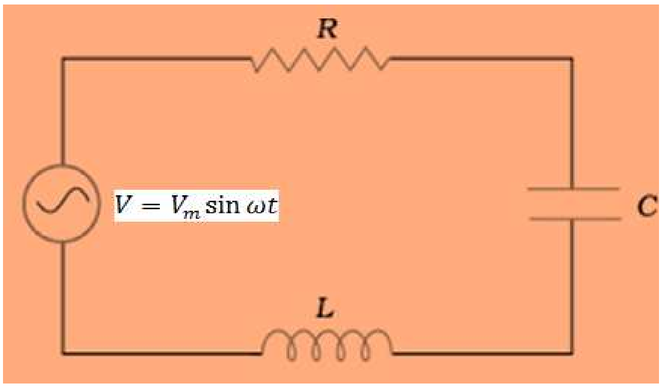
Series R-L-C circuit

Consider an A.C voltage,

$V = V_m \sin \omega t$.................. (1)

The resistor, inductor and capacitor are in series.

Therefore, **the ac current in each element is the same at any time**, having the same amplitude and phase.



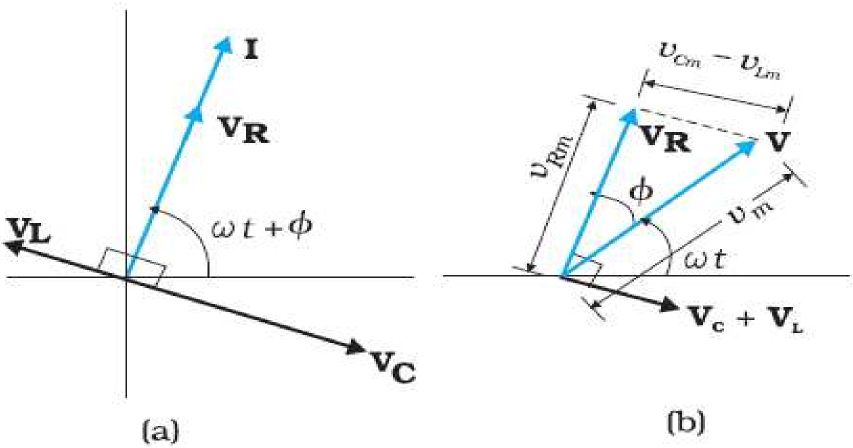
Let it be $I = I_m \sin(\omega t + \varphi)$ where φ is the phase difference between the voltage across the source and the current in the circuit.

(i) Maximum voltage across R $\Rightarrow V_R = I_m R$, In the resistive circuit voltage and current are in the same phase.

(ii) Maximum voltage across inductor $\Rightarrow V_L = I_m X_L$, In the inductive circuit current lags the voltage or voltage leads the current by π/2.

(iii) Maximum voltage across inductor $\Rightarrow V_C = I_m X_C$, In the capacitive circuit current leads the voltage or voltage lags the current by π/2.

Phasor Diagram for L C R circuit: -



From above figure (a) **I** represent the current in each element and $\mathbf{V_R}$**,** $\mathbf{V_L}$ **and** $\mathbf{V_C}$ represent the voltage phasors. $\mathbf{V_L}$ **and** $\mathbf{V_C}$ act along the same line but are in opposite directions.

$\mathbf{V_R}$ is along **I,** The difference phasor **(**$\mathbf{V_L + V_C}$**)** is perpendicular to the $\mathbf{V_R}$.

$Here, |\boldsymbol{V_L} + \boldsymbol{V_C}| = (V_C - V_L)$

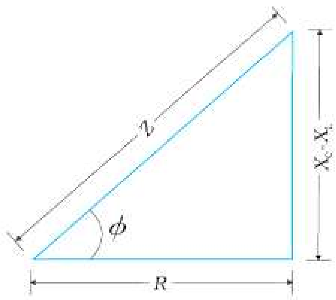
The Resultant Potential is given by $\boldsymbol{V} = \boldsymbol{V_R} + \boldsymbol{V_C} + \boldsymbol{V_L}$

Magnitude of Resultant Potential is given by

$$V_m = |\boldsymbol{V}| = \sqrt{{V_R}^2 + (V_C - V_L)^2}$$

$$V_m = \sqrt{(I_m R)^2 + (I_m X_C - I_m X_L)^2} = I_m\sqrt{(R)^2 + (X_C - X_L)^2}$$

$$Z = \frac{V_m}{I_m} = \sqrt{(R)^2 + (X_C - X_L)^2}$$



IMPEDANCE TRAINGLE

This is an effective resistance in series RLC circuit called the **Impedance Z.**

$$From\ \ figure, \qquad \tan\varphi = \frac{X_L - X_C}{R}$$

The impedance triangle for a series RLC circuit,

(i) If $X_L > X_C$ *then* $X_L - X_C$ is positive means voltage leads the current by phase angle of φ.

(ii) If $X_L < X_C$ *then* $X_L - X_C$= negative means voltage lags the current by phase angle of φ.

(iii) If $X_L = X_C$ *then* $X_L - X_C = 0$, means voltage is in the same phase with current and φ= 0.

Resonance: -

Resonance is said to be occur at this stage and the circuit becomes purely resistive.
When $X_L = X_C$, current has its maximum value and impedance has minimum value.

Maximum current at resonance

$$I_{max} = \frac{Z}{R}$$

Minimum impedance at resonance

$$Z_{min} = R$$

If frequency ω is varied, then at a particular frequency ω_0 , $X_L = X_C$ this is called **resonant frequency**.

$$\Rightarrow X_L = X_C \Rightarrow \frac{1}{\omega_0 C} = \omega_0 L \ \Rightarrow\ \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\nu_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

When $\nu_0 =$ natural frequency of the series LCR circuit becomes equal to frequency of AC source and current becomes the maximum then the phenomenon is called **Resonance**.

POWER IN AC CIRCUIT: THE POWER FACTOR

An A.C voltage, $V = V_m \sin \omega t$ applied to a series LCR circuit drives a current in the circuit given by $I = I_m \sin(\omega t + \varphi)$.

Instantaneous Power = Instantaneous Current x Instantaneous voltage, as voltage and current changes continuously in an ac circuit.

$$P = VI = V_m \sin \omega t \text{ X } I_m \sin(\omega t + \varphi)$$

$$Average\ power\ over\ the\ complete\ cycle = P_{av} = \frac{V_m I_m}{2} \cos \varphi = V_{rms} I_{rms} \cos \varphi$$

True power = apparent power or virtual power X cos ϕ

Power Factor = cos $\boldsymbol{\varphi} = \frac{\boldsymbol{R}}{\boldsymbol{Z}}$ = True power / Apparent power

(1) If ϕ =0°, cos 0°=1 ⇒ $Z = R$, means circuit consists of pure resistance.

(2) If ϕ =90°, cos 90°=0, ⇒ $0 = R \Rightarrow Z = X_L - X_C$ means circuit consists of pure inductance or capacitance or their combination in series.

(3) The average power consumed over a cycle is zero and current is called **wattless current**.

(4) If circuit has all the three elements R, L and C then

Power Factor = cos $\varphi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$.

From this we get the information that power spent in the circuit depends upon the power factor.

Transformer

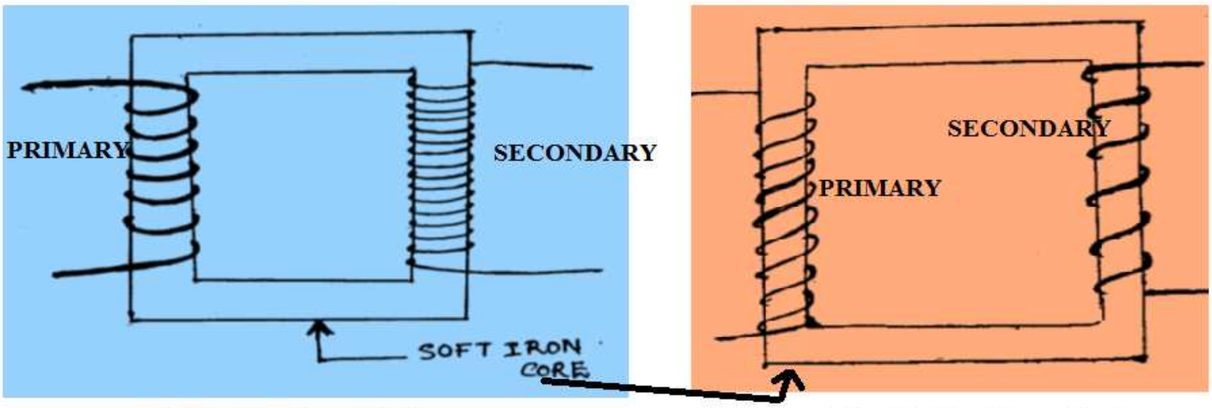
Transformer is a device used to increase or decrease A.C. voltage

Principal: - Transformer works on the principle of mutual induction.

Construction: -

When a.c. flows through the primary coil, a changing magnetic field is produced around it. The secondary coil is placed in this changing magnetic field and hence an e.m.f. is induced across the secondary coil.

Step Up Transformer **Step Down Transformer**



$N_s > N_p$, $V_s > V_p$ and $i_s < i_p$ **$N_s < N_p$, $V_s < V_p$ and $i_s > i_p$**

Let **Np** and **Ns** be the number of turns in the primary and secondary of the transformer. The voltage induced in the secondary, when **AC** flows through the primary is given by

$$\mathbf{Vs = Ns}\frac{d\phi}{dt} \text{.......... (1)}$$

At the same time, due to self-induction, the back e.m.f. produced in the primary is

$$\mathbf{Vp = Np}\frac{d\phi}{dt} \text{......... (2)}$$

Dividing equation (1) by (2)

$$\frac{\mathrm{Vs}}{\mathrm{Vp}} = \frac{N_s}{N_P} = \frac{I_P}{I_S} = k$$

Efficiency of transformer,

$$\eta = \frac{\text{output power}}{\text{input power}} X100\%$$

A.C. Generator

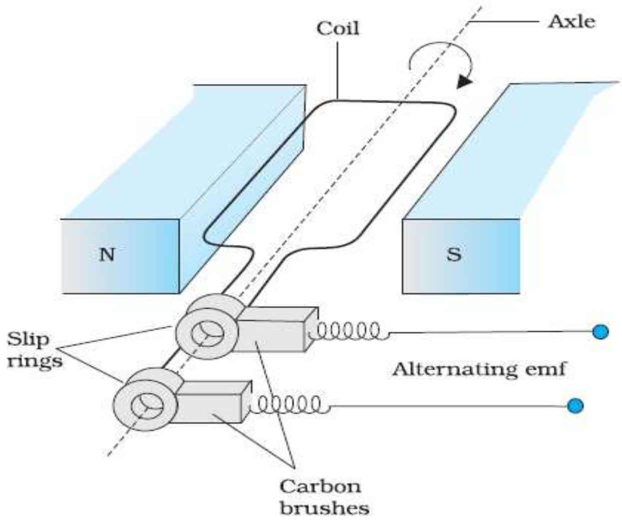
It is a device which convert mechanical energy in to an electric energy and generates alternating current.

Principle

A.C. generator works on the principle of electro-magnetic induction.

Construction: -

1.Armature coil
2.field magnet
3.slip ring
4.Brushes



A C GENERATOR

Theory: -

When the armature coil rotates between the pole pieces of field magnet,

The effective area of the coil is A cos θ,

The flux at any time is $\phi = \vec{B}.\vec{A}$ = NBA cos θ

where θ is the angle between Area vector $\vec{A}$ and magnetic field $\vec{B}$.

Since, coil is being rotated about its axis to change the flux ϕ at rate ω rad per sec.

$$\phi = NBA \cos \omega t$$

Where ω is the angular frequency of the coil.

The induced e.m.f.

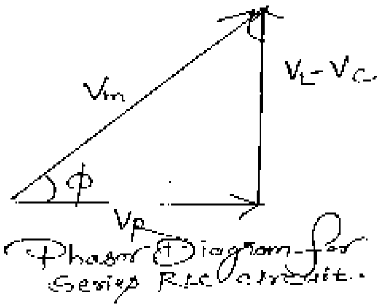
$$\varepsilon = -\frac{d\phi}{dt} = -N\frac{d(BA\ \cos\omega t)}{dt} = -NBA\omega \sin\omega t$$

$Magnitude\ of\ induced\ EMF \quad V = \varepsilon = NBA\omega \sin\omega t$

V_{max} = NABω = V_0 (numerically), which is the maximum value of induced emf.

Formulae: -

1. $v = v_m \sin\omega t = v_m \sin 2\pi\nu t$ and $i = i_m \sin 2\pi\nu t$
   where, i and v are the instantaneous values and i_m and v_m are the peak values.
2. $i_{rms} = \frac{I_m}{\sqrt{2}}$ rms, virtual or effective values of ac current.
3. $v_{rms} = \frac{V_m}{\sqrt{2}}$ rms, virtual or effective values of ac voltage.



4. $X_L = \omega L$ inductive reactance.
   Unit: - ohm Ω
5. $X_C = \frac{1}{\omega C}$ Capacitive reactance.
   Unit: - ohm Ω
6. In an ac circuit containing inductor current lags or voltage leads by π/2.
7. In an ac circuit containing capacitor current leads or voltage lags by π/2.
8. $Z = \sqrt{R^2 + (X_L - X_C)^2}$ Impedance of series RLC circuit.
   Unit: - ohm Ω.
9. $\tan\phi = \frac{X_L - X_C}{R}$ where φ is the phase angle between current and voltage.
10. $P_{av} = v_{rms} i_{rms} \cos\phi$
11. Unit: - W (watt)
12. $P_{av} = i_{rms}^2 R$ ……. equation obtained from above phasor diagram.
13. Power factor = $\cos\phi = \frac{R}{Z}$, in case of an ideal capacitor or a capacitor $\phi = 90°$ and $P_{av} = 0$
14. $i_m \sin\phi$ = wattless component of current.